MSG No. 90.100 P_c42_12 [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.22`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,3,4,5]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[9,14,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,12,13]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,6,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,3,12,13]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,5,10,11]`
4	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `4..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,3,6]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[4,5,7,8]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[12,13,15,16]`

Table 4: Wyckoff site: `8d`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5]`
4	$[0,\ 0,\ -z]$	`[7,8]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13]`
8	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[15,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[x,\, x,\, \frac{1}{2}\right]$	[9,15]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[10,13]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[11,12]
8	$\left[-x,\, -x,\, \frac{1}{2}\right]$	[14,16]

Table 6: Wyckoff site: 8f, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[2,13]
3	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[3,12]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[4,11]
5	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[5,10]
6	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[6,16]
7	$\left[x,\, x,\, \frac{3}{4}\right]$	[7,9]
8	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[8,14]

Table 7: Wyckoff site: 16g, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[2]
3	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[3]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$[y,\, x,\, -z]$	[7]
8	$[-y,\, -x,\, -z]$	[8]
9	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[11]
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[15]
16	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[16]